

CUAS Astrophotography Guide



2026 Edition

In 2026, CUAS obtained a new telescope and related equipment, specifically intended for deep-sky astrophotography. This document explains how to use it.

This guide assumes knowledge of basic astronomy terms, most of which should be familiar after using the Northumberland telescope, and any others can be easily googled.

[Quick Start](#) contains a sequence of steps that should be followed to perform a complete photography session, assuming everything has been left correctly. [Astrophotography Basics](#) has brief explanations on what is necessary for astrophotography in general, since this can be quite a complex topic. [Our Equipment](#) contains details and manual links for all of our equipment, with some details on setup when things aren't quite as they should be.

Please contact us at cuasobserv@gmail.com if you have any questions, problems, accidents, or notice anything out of the ordinary. This will help us stay on top of the equipment status, and keep it available for members for many years to come.

Quick Start	2
Astrophotography Basics	7
Our Equipment	9
Admin	12



Cover photo: Markarian's Chain, taken from the Northumberland Lawn over several nights, with the equipment in this guide.

Quick Start

To use the camera, you must have the ZWO ASIAIR app installed on your phone or tablet, or on a laptop via BlueStacks (an Android Emulator) if you prefer. **Please install ASIAIR before coming for a training session.**

Follow the following steps to do a complete astrophotography session.

I'll emphasise here to be extremely careful with all parts of the telescope system, since anything broken stops not only you, but everyone else, from being able to use it.

If any unexpected accidents do happen while using it, please let us know even if there isn't any damage. This will help us to train people on it better.

Please be considerate of others using the area, particularly keeping torches off and screens at their lowest brightness, to maintain both their night vision and yours. I use a red bike light if there is something I need to see.

While moving and using the telescope, be as careful as possible to not let excess dust into the optics. This can be done by keeping the lens cap on whenever possible, and by not removing the camera where not necessary.

When leaving the telescope to take photos, keep an eye on it and **DO NOT** let it get rained on. Keep an eye on both the forecast and the sky, and do not use it if rain is likely.

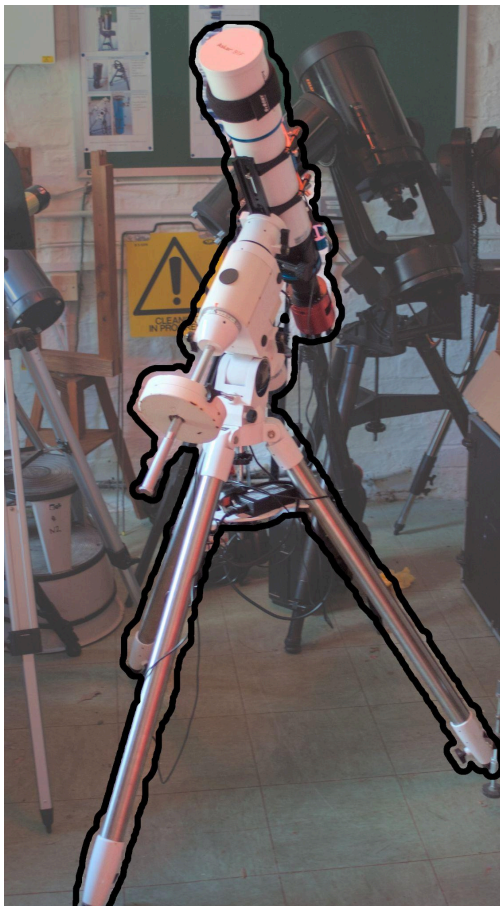


Fig. 1

The telescope's location in storage.

Steps:

1. Enter the Northumberland as usual, disable the alarm and sign into the logbook (Write 'Askar' instead of 'N' in the Telescope column).
2. The astrophotography telescope is found in the room to the south of the Northumberland Telescope corridor, which can be opened with the same key as the building. Enter and check that the telescope is assembled as seen in Fig. 1. If it is not, refer to the [Our Equipment](#) section to assemble the telescope, and/or contact us if you don't feel confident setting it up.
If any accessories are required, these are stored in the boxes in the top shelf of the cabinet in the main Northumberland room.
3. Making sure the telescope is not plugged in, and no cables are going to catch on anything, lift the entire mount gently outside.
You must only move it by holding the mount, not the optical tube. Be certain neither end of the telescope hits either of the door frames. This may require two people.
4. Place the mount on the grass near one of the two power stands, making sure the 'N' on the mount is facing roughly north (Fig. 4). It's worth learning the position of Polaris in the sky to be able to do this quickly. Adjust the mount legs so that the mount is horizontal.
5. Remove the power stand cover, and plug the camera in. This will feed through to power the mount, which must be switched on.
6. The mount must be polar aligned in order to accurately track the stars. My procedure is as follows - refer to Fig. 2 for mount control descriptions:
 - Remove both caps to the mount's polar scope.
 - Disengage the mount's DEC axis, rotate the tube to a declination of 0, and reengage the DEC axis. This allows light to pass down the polar scope.
 - Look through the polar scope. You should see a red constellation diagram (Fig. 5) and, if the alignment in step 4 was sufficient, the single bright star Polaris. If you don't see Polaris, rotate the entire tripod until you do.
 - Disengage and rotate the RA axis to rotate the red diagram to match the constellations seen in the sky. Be careful here to not collide the tube or camera with the mount here, and lock RA after this.
 - Adjust the alignment using the two pairs of push-pull bolts to move Polaris into the small circle in your rotated diagram. This is necessary since Polaris is not precisely on the celestial pole.
 - Rotate the tube back, and replace both polar scope caps.There are other polar alignment methods involving the ASIAIR app or the official SkyWatcher mount guide, described in their respective manuals ([Our Equipment](#)) which may be useful if you require more precision.
7. Finally, remove the telescope lens cap, make sure the dew shield is extended, and turn the dew heater on. You are now ready to connect to the camera.

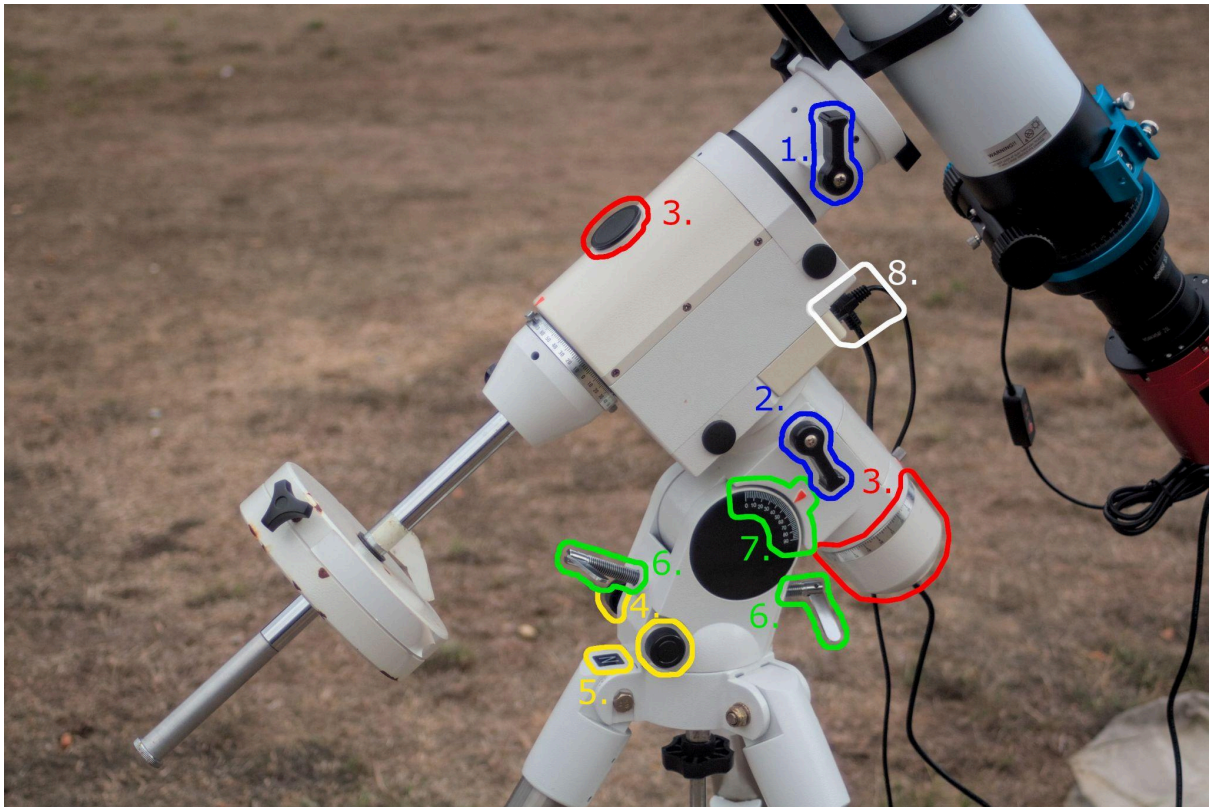


Fig. 2 - Annotated photo of the mount, with relevant parts highlighted.

1. Declination lock - loosen to rotate freely in DEC, and tighten afterwards.
2. Right ascension lock - loosen to rotate freely in RA, and tighten afterwards.
3. Lens caps for the polar scope.
4. Conjugate screws for adjusting Azimuth during polar alignment.
5. Large deviations from North should be corrected by rotating the entire mount instead.
6. Conjugate screws for adjusting Altitude during polar alignment.
7. Large deviations from Cambridge's latitude of 52°N should be corrected by adjusting the tripod feet length instead.
8. The mount requires both a power and a control cable from the camera, here.

A detailed guide and manual for the camera can be found in the [Our Equipment](#) section, but the important parts are covered below. I will not cover every button press required, only the gist of what must be done to take photos. It's a bit fiddly, so it may take a while to get the hang of the app and get it working for the first time, and may require referring to the ZWO guide for details and locations of menu items.

8. Open the ASIAIR app, and connect your device to the camera's wifi network, 2600AIR_45902005, with password 12345678.
9. You will be prompted to enter the telescope focal length. If there is a focal reducer attached, enter 455mm for both the main and guide cameras, otherwise enter 609mm. For more details on the focal reducer, see [Our Equipment](#).



Fig. 3 - The rear of the telescope, showing the optical path to the camera. Note that this is with the focal reducer installed, and may look slightly different otherwise.

1. Focuser, with knobs for coarse (black) and fine (blue) focus adjustments.
 2. Rotation wheel and locking knob.
10. Make sure the mount is connected and tracking is switched on, in the mount tab of the top menu.
 11. Switch to Focus mode in the right panel, and adjust the exposure setting until you can see stars. Adjust the focus knob on the telescope (Fig. 3) until the stars are as small and sharp as possible. The app can numerically evaluate the focus, which may help achieve an optimum.
 12. The camera's sky model must now be calibrated to the real sky. Make sure the tube is pointing further than 10° from the celestial pole and meridian, by disengaging an axis moving it manually, and reengaging it.
 13. Switch back to Preview mode, set the exposure for 5s, and take a photo. Platesolve this image using the tool on the left panel, and sync the mount.
 14. Slew the telescope to your desired object by using the 'Tonight's best' menu, by searching for an object, or by panning to the location from the starmap in the bottom left.
MAKE SURE the telescope is slewing in the correct direction; it can get confused and require a new platesolve or a complete camera reboot. If it is slewing incorrectly (e.g. so as to point at the ground), stop it immediately otherwise the camera could collide with the mount.

15. Take a preview exposure to check required exposure time and framing. You may want to rotate the camera using the ring on the telescope (Fig. 3), but this would necessitate repeating step 13.
16. Once the object is correctly framed, you can switch on guiding from the left panel. Tap the guiding window that appears, adjust the exposure (1-2s works well), and start guiding. The app should find some stars and start filling in the accuracy graph. This should stabilise to an error of 0.5-2.0", depending on seeing and polar alignment. If it doesn't, then the guide camera may need recalibrating which takes a few minutes.
17. Return to the main screen and switch to Autorun mode. Add a 'Lights' sequence with reasonable exposure time for your object, and a sensible number of frames. To make your images compatible with our pre-taken Dark and Bias calibration frames, your frames should:
 - Have an exposure time of 20, 60, 180, 300, or 600 seconds.
 - Have a gain of 100.
 - Have a camera temperature of -10°C.Enable Meridian Flip if your object will cross the meridian during your shooting session.
18. Start the sequence, and wait for at least the first image to make sure it all looks good. You must stay nearby to keep an eye on the telescope while it is imaging, to make sure it doesn't collide into itself or get rained on, and it must be packed up by dawn.
19. Once you are done, shut down the camera from the top menu and unplug the power. Replace the telescope cap and the cover to the power stand. Move the mount such that the telescope is pointing at DEC +90 and parallel to the meridian. Carry the entire telescope back inside, holding it only by the mount as before.
20. Any images taken are stored in the camera itself. To transfer them to your own device, use a USB-C cable or memory stick in the back of the camera, and move the files across through the ASIAIR app. Images are stored in raw (.fits) format in the Autorun folder. Please only remove your own images, since someone else may not have saved theirs yet.
21. You will need to take flat calibration frames (explained in [Astrophotography Basics](#)) before or after your main frames, without changing the optical path at all. This means no refocussing, no camera rotation and no dust added or removed. The required flat illumination can be a bit tricky; I've had luck by returning the next day and pointing the telescope at the ceiling of the storage room, lit by light from the window. A (dust-free!) t-shirt over the end of the tube helps to diffuse the light further. Take around 30 frames using a Flat schedule within Autorun.
22. The telescope can be left either with or without the focal reducer installed. Note the camera angle (Fig. 3) if planning to return to the same object another night.

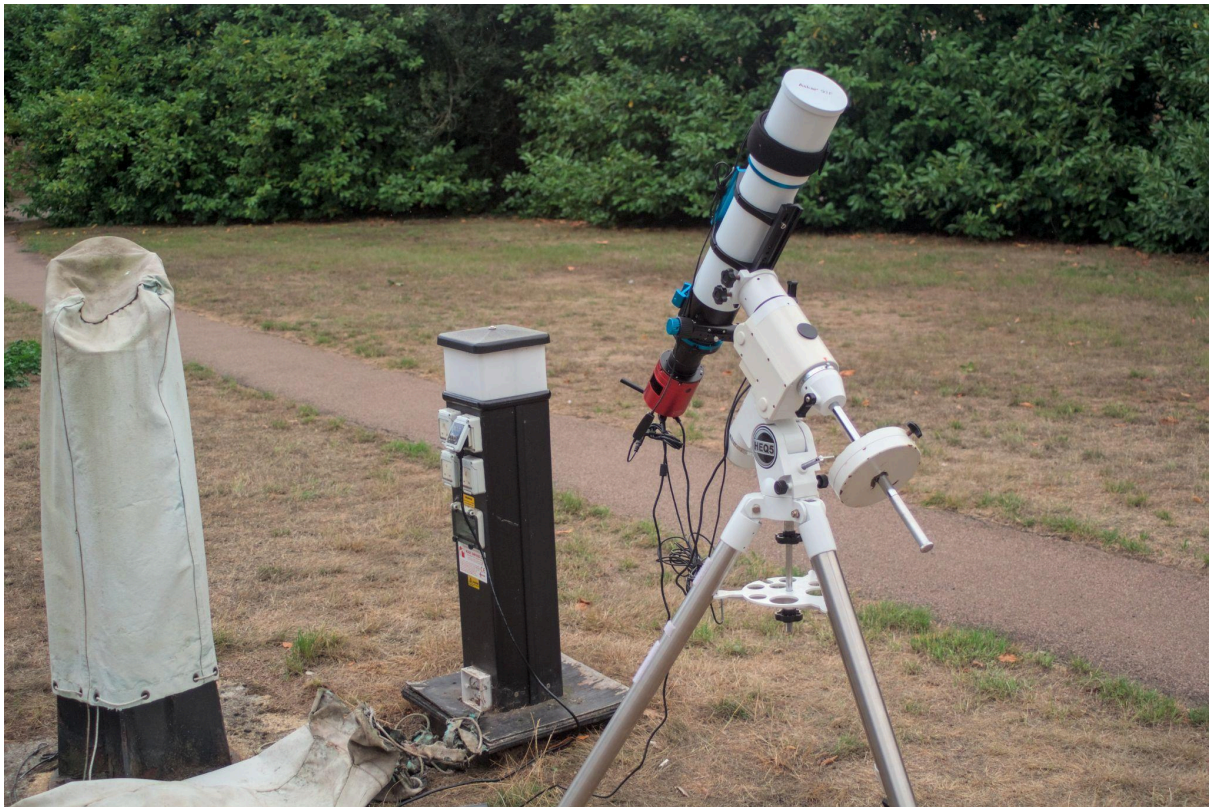


Fig. 4 - The telescope set up outside and ready to image once it's dark. Note the power outlet location and the rough northwards orientation of the telescope.

Astrophotography Basics

Deep-sky astrophotography is the process of imaging incredibly faint things. While the cores of brighter objects can be seen in short exposures, the process of bringing out faint background detail, reducing noise and creating a beautiful final image require quite a lot of care. This section briefly explains some of the basic techniques.

There is a lot of good information online to further your knowledge; I've found <https://astrobackyard.com/> and the forums at <https://www.cloudynights.com/> really good for explanations.

To make a good astrophoto, you need a few things:

- **Object Choice.** Our system is wide-field, low focal length, which means certain objects will give better results than others. Emission nebulae, star fields and large galaxies will nicely fill the frame. Smaller galaxies and nebulae might need significant cropping and won't give a wallpaper-worthy image. Planets, planetary nebulae and the smallest galaxies won't show much detail at all, and really need a longer focal length to do them justice.

To help plan framing, you can use a field of view calculator, such as https://astronomy.tools/calculators/field_of_view/. Our setup gives a view of $2.21^\circ \times 1.48^\circ$ in the default configuration and $2.96^\circ \times 1.98^\circ$ with the focal reducer.

- **Exposure time.** Without this, there will be no or only noisy data about the faint detail in your image, whatever you do afterwards.
This will take multiple nights on the same target to get photos comparable to the best amateur ones seen online.
- **Stacking.** You cannot have the camera exposed for hours continuously, since stars and eventually the sky background will saturate the sensor, tracking errors will be magnified into wiggly stars, and satellite/plane trails will saturate the image.
Instead, take many short exposures of a few minutes each (short enough that only the brightest stars are saturated). These can be aligned and intelligently 'stacked' in software later.
- **Calibration frames.** Any photos taken will have a combination of errors on top of the data. Lots of this error can be removed with carefully prepared calibration frames.
 - **Darks:** photos taken for the same length of time as the Light frames, but with the lens cap on. This captures fixed pattern 'dark current' noise, which can be subtracted from the image during stacking.
 - **Flats:** short photos of an evenly lit background. This shows dark spots where dust partially obscures the camera sensor, as well as dark vignetting around the edges. Stacking software will divide your final image by this to get an even illumination.
 - **Bias:** short photos with the lens cap on. These are used to correct the flats for any fixed pattern noise. They are probably almost unnecessary given how good modern CMOS technology is in our new camera.

For each type of calibration frame, take around 30 and stack them to reduce shot noise.

Dark and Bias frames are typically valid for months or years, until the sensor degrades significantly. Therefore, we've taken frames with the following settings

- An exposure time of 20, 60, 180, 300, or 600 seconds.
- A gain of 100.
- A sensor temperature of -10°C.

These can be freely used and be accessed here:

https://drive.google.com/drive/folders/1GsJiUcl-ZGun_kuXxh2G_vQnbcvZEWqU.

Let us know if they get out of date and we'll prepare new ones.

Flats must unfortunately be taken every session, and suggestions are described in the [Quick Start](#) section

- **Tracking.** To achieve sharp pointlike stars, your mount must point the telescope at the same part of the sky, with a precision of 1-2 arcseconds. This is done with two main techniques.
 - Polar alignment. The equatorial mount rotates the RA axis at the sidereal rate to follow the stars as the earth rotates under them. This requires the axis to be perfectly aligned with earth's rotation axis. The procedure for this is described in the [Quick Start](#) section.
 - Guiding. Polar alignment is never perfect, and there are other slight long period errors in tracking that can be corrected for. Guiding uses a separate

camera to monitor star position in real time, and give tiny adjustments to the mount to keep them in the same place.

Our camera has two sensors, one of which is a guide sensor which makes setting up guiding very easy - it uses the same telescope, which is fairly unusual. This can be set up in the app as described in the [Quick Start](#) section.

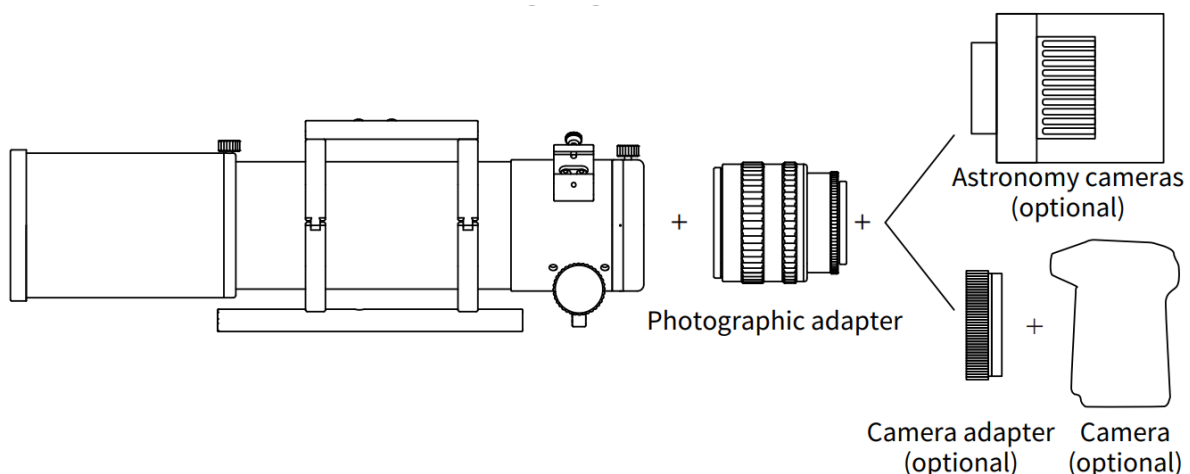
- **Processing.** To bring out all the detail in the image, a fair amount of digital processing is needed, including stacking, background removal and stretching. While this is beyond the scope of this guide, here are a few tools to get you started:
 - DeepSkyStacker: Free stacking software, very simple to use. It combines all your Light and Calibration frames into a Master image that can be edited elsewhere.
 - Graxpert: Free software for removing the light-pollution induced background and gradients from your stacked master, and can do stretching as well.
 - Siril: The best free astrophotography software I've used, it handles stacking and some post-processing. It can be scripted and can spit out a complete final image, at the cost of a bit of a learning curve.
 - PixInsight: State-of-the-art paid software, worth buying and learning if you are serious about the hobby.

I'll point out that astrophotos must have a non-linear stretch applied for faint detail to be visible, otherwise everything will look black compared to how bright the brightest stars are.

Our Equipment

Askar 91F Flat-Field Refractor. This is the 91mm aperture, 609mm focal length F6.7 telescope, which acts to gather lots of starlight and focus it onto the camera sensor. It includes the focussing mechanism, as well as attachment points to the mount and camera. The manual is available at https://www.sharpstar-optics.com/download_Detail_1/63.html.

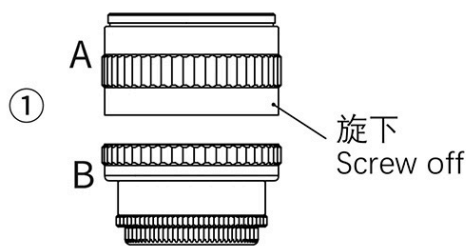
Of note is the diagram below of the optical path when attaching to a camera:



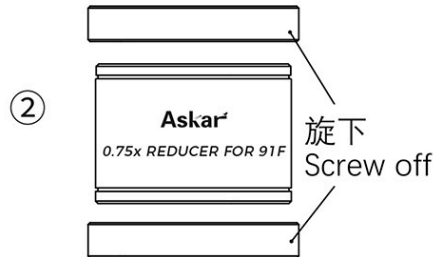
Askar 0.75x Reducer for 91F: This device changes the focal length of the telescope to 455mm and reduces the focal ratio to F5. This essentially provides a larger field of view and

more light focussed per camera pixel (so less exposure time is needed). This comes at the cost of less magnification and fine detail.

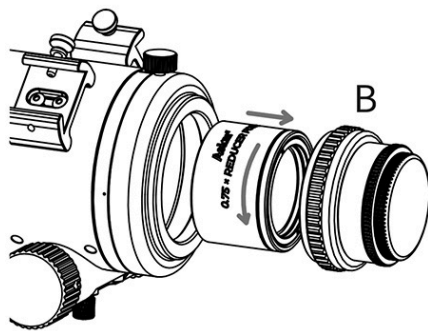
To attach, follow the diagram below. To remove, do the opposite, put the reducer carefully away, and reattach the camera as shown in the diagram above. **BE CAREFUL** with any screw threads, since they are very fine and can be irreversibly damaged if the wrong angle and/or any force is used when attaching.



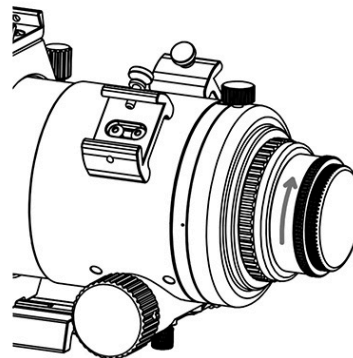
① 旋下91F拍摄延长管A
Screw off the 91F photographic extension tube A



② 旋下0.75x减焦镜两侧端盖
Screw off the end caps on both sides



③ 按照箭头方向把0.75x减焦镜
旋入延长管B
③ Attach the 0.75x reducer to
extension tube B in
the direction of the arrow



④ 按照箭头方向旋入91F即组装完成
④ Attach the 0.75x reducer to the
91F in the direction of arrow, and
the assembly is complete.

Sky-Watcher HEQ5 Mount: This is the equatorial mount and integrated tripod which holds and balances the telescope. It has two motorised axes which are entirely controlled by the camera, and a polar scope for aligning it to the celestial sphere. A description of commonly used knobs is shown in Fig. 2.

The manual is available at

https://inter-static.skywatcher.com/upfiles/en_download_caty01461887945.pdf

It should be always kept in the correct state, but there are a few things useful to know which may need adjustment when setting up:

- The height of the tripod feet may need to be adjusted to get the mount horizontal. I judge this by eye, looking at the flat spreader plate.
- The balance may need to be adjusted by sliding the telescope on its bracket to balance DEC, and sliding the counterweight on its shaft to balance RA. A balanced

telescope puts much less strain on the mount, allowing better tracking and less wear and tear.

- When polar aligning using the two pairs of push-pull bolts, one bolt **MUST** be loosened before the other is tightened, otherwise the mechanism will be damaged. Never tighten these bolts too much.

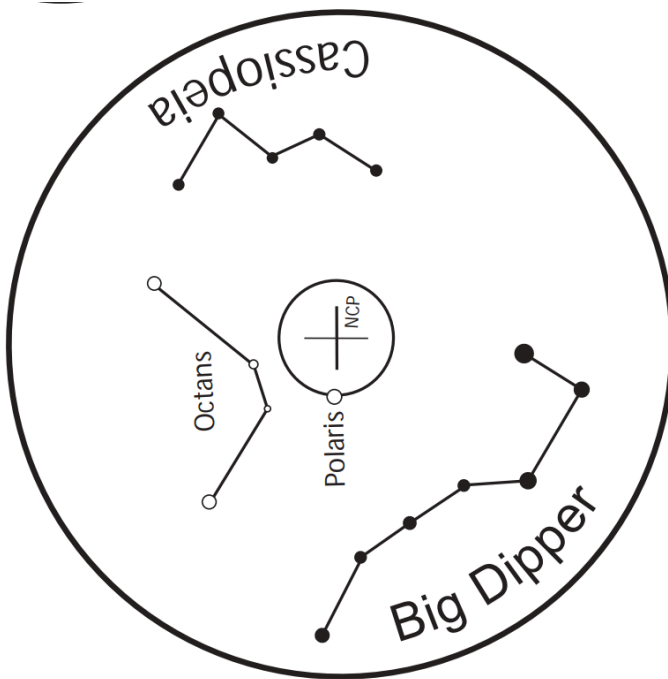


Fig.5 - Diagram of the view through the HEQ5's polar scope, showing where Polaris must be relative to the Big Dipper and Cassiopeia. (Octans is only relevant in the southern hemisphere).

ZWO ASI2600MC Air: This is our all-in-one smart one-shot-colour (OSC) camera. It has two sensors, including an 26 megapixel APS-C imaging sensor and a smaller guide sensor. It is a cooled camera, which can reduce the sensor temperature to 30-35°C below ambient, which significantly reduces dark current noise.

It contains a mini computer and is entirely controlled wirelessly via a dedicated Wi-Fi network and the ASIAIR app. It interfaces directly with the mount, and has feed-through power to the mount and dew heater, so the entire setup can run off one plug.

The manual is available at

<https://i.zwoastro.com/wp-content/uploads/2025/11/fc9b30ae6ea0aa4b331322439dee063c.pdf>

Svbony SV12 Dew Heater: This slips around the dew shield of the telescope, and is powered through the camera. To turn on, press the button on the cable. It slightly warms up the objective lens to above the dew point, preventing droplets forming and scattering light.

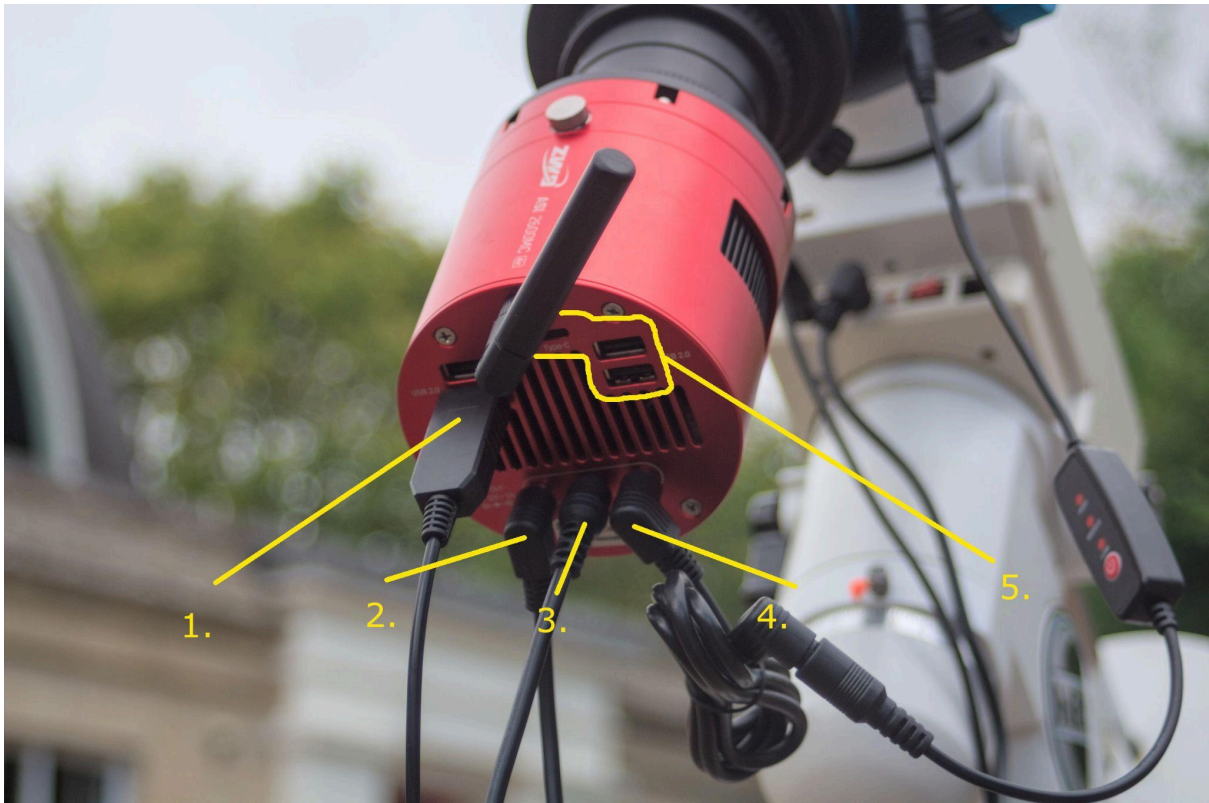


Fig. 6 - Annotated photo of back of the camera, showing necessary cables:

1. Control cable for the mount.
2. Power cable from camera to the mount.
3. Power cable for the camera; this feeds power through to the other devices.
4. Power cable from camera to the dew heater.
5. Empty connections usable for file transfer after imaging.

Admin

To use the equipment described in this guide, you must:

- Be a member of CUAS.
- Have been to two Northumberland telescope ObsDemos and passed an ObsTest.
- Be sure that you are interested in astrophotography enough to pursue this hobby. It will take quite some independent effort to learn it from scratch, and it uses a lot of our time to train people who do not put their own time in and don't end up using the equipment.
- Sign up for the astrophotography training waiting list.
- Have read this guide in its entirety, and installed the ASIAIR app.
- Sign up to a one-on-one nighttime training session, where one of us will supervise you setting up the equipment and answer any questions. If we deem you ready and sufficiently careful, you will be allowed access to the equipment.

If you wish to use the camera on the 16-inch (if you have access) or Northumberland telescopes, either ask at the training session or email us at cuasobserv@gmail.com.

There is no Booking System for the astrophotography telescope and is used first-come-first-serve, at the time of writing.

Astrophotography DataShare

Users of the Astrophotography Equipment are encouraged to upload stacked masters and/or final edits to the CUAS Members Astrophotography Data Share available here:

<https://drive.google.com/drive/folders/1JGg1PguuxeW7vDbh9FeBoO4hbsnu1LW4?usp=sharing>

Please create a fresh subfolder in this Data Share for each Astrophotography Target, and include your name in the subfolder name.